WETLAND DEGRADATION TECHNICAL REPORT

Multi-Temporal Spectral Transition Matrix, CVA Displacements, NDWI Dynamics & Fusion

RUN ID	run_20260929_115746	SOFTWARE VERSION	1.0.0
ANALYSIS MODE	wetland (two-date temporal)	**EXECUTION TIMESTAMP**	2026-09-29T11:58:29 UTC
OBSERVATION PERIOD	2023-08-14 → 2023-08-28 (14 days)	**ENVIRONMENT**	Python N/A | N/A

1. TECHNICAL OVERVIEW & GRID ALIGNMENT

GRID SPECIFICATION	VALUE
Spatial Reference (CRS)	EPSG:32633
Reference Grid Dimensions	100 rows × 100 cols
Pixel Resolution	10.00 m × 10.00 m
Bounding Extent [W, S, E, N]	[500000.00, 4999000.00, 501000.00, 5000000.00]
Analysis Footprint Area	35.00 ha (3,500 valid pixels)
Wetland Boundary Source	Project-provided wetland boundary

2. POST-CLASSIFICATION TRANSITION ANALYSIS

Every valid pixel is classified into one of 16 discrete state transitions between T1 and T2. Classes: 0=Background (B), 1=Severe Stress (S), 2=Moderate Stress (M), 3=Healthy (H):

TRANSITION CATEGORY	PIXELS	AREA (HA)	SHARE (%)	ECOLOGICAL INTERPRETATION
Stable Landscape	3,500	35.00	100.00%	Canopy condition remained unchanged between observation dates.
Vegetation Degradation	0	0.00	0.00%	Transition from higher to lower health state (H→M, H→S, M→S).
Vegetation Recovery	0	0.00	0.00%	Transition from lower to higher health state (S→M, S→H, M→H).
Canopy Loss (Veg → Bg)	0	0.00	0.00%	Canopy disappearance into soil, water, or bare surface.
Emergence (Bg → Veg)	0	0.00	0.00%	Vegetation emergence over formerly bare or water surface.

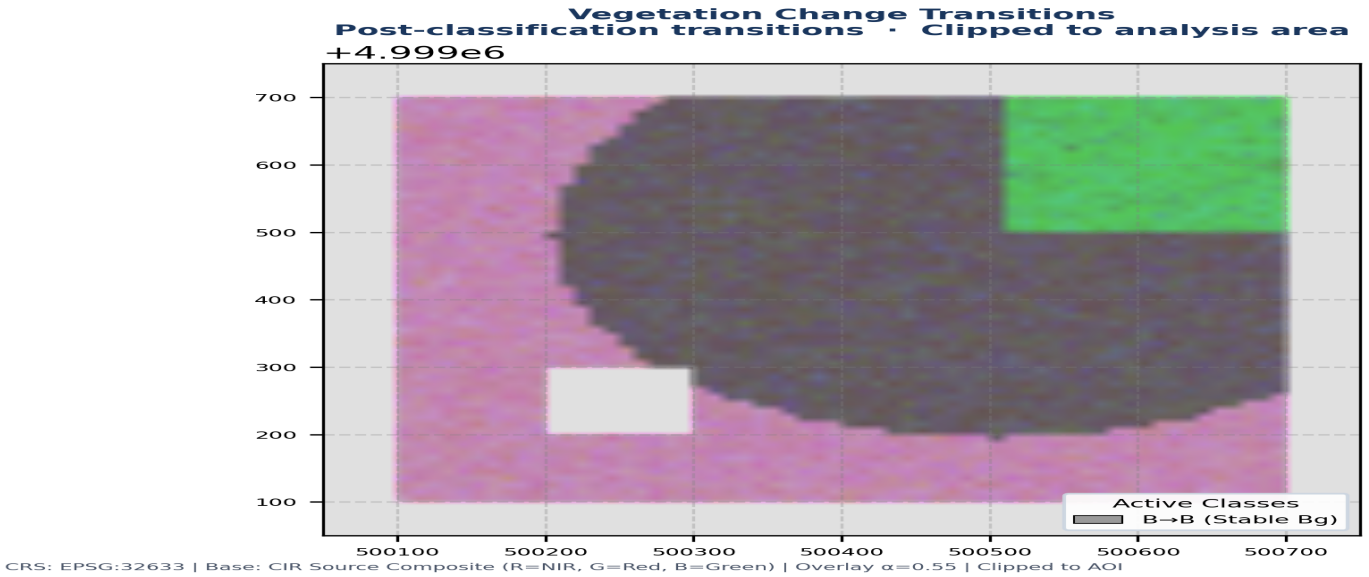


Figure 1: Classified 16-State Transition Cartographic Plate

3. CONTINUOUS CHANGE VECTOR ANALYSIS (CVA) & SPECTRAL SHIFTS

METRIC	FORMULA / DESCRIPTION	VALUE
Mean CVA Magnitude ‖ΔV‖	`sqrt(ΔNDVI² + ΔGNDVI² + ΔNDRE²)`	0.6969
Median CVA Magnitude	50th percentile of magnitude surface	0.6719
95th Percentile Threshold	P95 statistical outlier threshold	1.3386
Mean ΔNDVI	`NDVI(T2) − NDVI(T1) (canopy vigor shift)`	+0.4944
Mean ΔGNDVI	`GNDVI(T2) − GNDVI(T1) (chlorophyll shift)`	+0.2689
Mean ΔNDRE	`NDRE(T2) − NDRE(T1) (red-edge foliar shift)`	-0.0334

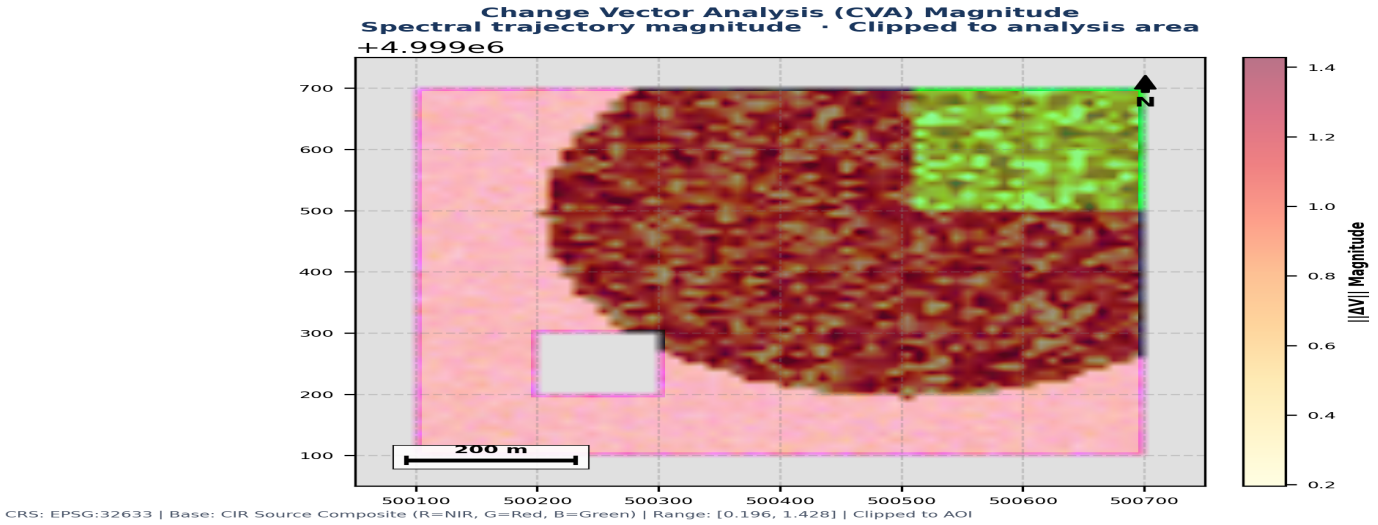


Figure 2: Continuous Change Vector Analysis (CVA) Magnitude Plate

4. HYDROLOGICAL EVIDENCE & SURFACE WATER DYNAMICS

HYDROLOGICAL PARAMETER	VALUE
NDWI Water Threshold	`0.00 (Green − NIR) / (Green + NIR)`
Surface Water Contraction (Loss)	0.00 ha
Surface Water Expansion (Gain)	3.78 ha
Net Water Surface Change	+3.78 ha

5. MULTI-EVIDENCE FUSION & DEGRADATION ASSESSMENT

ASSESSMENT INDICATOR	SCIENTIFIC METRIC	STATUS / VALUE
Hydro-Ecological Degradation	0.00 ha	Dual confirmation: vegetation degradation + surface water drying.
Hydro-Ecological Recovery	0.00 ha	Dual confirmation: vegetation recovery + surface water expansion.
Normalized Degradation Ratio (NDR)	0.0000	Degraded canopy area normalized against total valid baseline vegetation.
Severity-Weighted Degradation	N/A	Weighted sum of transitions (H→S weight 1.0, H→M weight 0.5, M→S weight 0.5).
Physical Wetland Depletion	NOT_ASSESSED	Conservative safeguard: only confirmed when independent ground/boundary evidence corroborates.

6. QUALITY ASSURANCE & PROVENANCE AUDIT

AUDIT COMPONENT	STATUS	AUDIT DETAILS
Mathematical Reconciliation	PASS	Transition cell counts sum exactly to effective footprint (error < 0.001%).
Spatial Alignment QA	PASS	T1 and T2 raster footprints aligned to common grid without rotation errors.
Radiometric Integrity	PASS	CVA and spectral indices checked for numerical stability and inf/nan absence.